

Linux Desktop Experience with ARM and MIPS

The Linux experience for many of our readers most likely starts with an x86 hardware platform. However, desktop solutions exist for other platforms, like ARM and MIPS.



When Linus Torvalds' practice with Minix led to Linux, the main development platform was a PC with the i386 architecture. Three years later, in 1994, he was asked to help Digital (DEC, or just Digital Equipment Corporation) port Linux onto the Alpha platform. After roughly a year, in 1995, it was officially declared that Linux supported two architectures—x86 and Alpha. However, unofficially, there was support for systems based on Motorola 68K as well.

At the beginning of the 2000s, the Linux source tree had hardware support for a dozen architectures: x86, Alpha, Itanium, MIPS, SPARC, ARM and others. While several were designed to be used inside big crunching systems (Itanium, Alpha and SPARC), others like ARM, and the vast majority of MIPS, were focused on energy efficiency. Thus, almost all mobile phones today run ARM processors, just as tablets from Apple, Samsung, Acer and Lenovo. While one can find MIPS CPUs in the mobile segment, it isn't widespread yet, because MIPS processors have conquered, and are concentrated in, various set-top boxes, DSL modems, etc. Surprisingly, you might find MIPS in several desktops and networks.

Today, we take a look at two systems with non-x86 CPUs: a TrimSlice H250 net-top (ARM-based) and a Yeeloong 8089 netbook (MIPS). The first is manufactured by CompuLab with software support provided by Canonical and Linaro, which made it possible to run Ubuntu for ARM systems. The second, manufactured by Lemote, uses the Debian MIPSEL OS. Additionally, do note that Lemote isn't the only driver for porting software to MIPS. For example, the FSF fund is actively involved in the development of the Linux MIPS port, making possible the gNewSense distro.

Ubuntu at TrimSlice (ARM)

Canonical and Linaro's joint efforts for Ubuntu on ARM means that you can find Ubuntu running on the Texas Instruments OMAP3/OMAP4 (BeagleBoard XM), Freescale i.MX5 (Efika MX smart-book), Nvidia Tegra2 (Toshiba AC100, TrimSlice) as well as a server appliance known as Project Moonshot, which is based on Calxeda ARM technology chips. What is so remarkable about ARM? First, its extremely low power consumption—about 1.5 watts per CPU. Second, its chip size is much smaller than competing products. Third, there's an integrated video chip built into the main CPU core. This symbiotic architecture offers a relatively cheap, but fairly efficient solution, because multimedia streams are decoded by specialised hardware video chips. True, you might not want to use it for a server with heavy loads (except ARM server appliances, where dozens of ARM CPUs work together), but as a desktop system for text editing, Internet browsing, and listening to music.

Hardware performance

Let's find out what this TrimSlice H250 net-top has:

```
trin@trimslice:~$ lspci
00:00.0 PCI bridge: nVidia Corporation Device 0b00 (rev a0)
01:00.0 Ethernet controller: Realtek Semiconductor Co., Ltd.
RTL8111/8168B PCI Express Gigabit Ethernet controller (rev 03)
```

```
trin@trimslice:~$ free
              total        used        free      shared    buffers
cached
Mem:           896120      879248       16872           0        7092
459192
```